WORKED SOLUTION

STEP III 1999, Question 3

Pure • Riemann Sums and Limits

STEP III 1999, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q3

The key idea

The expression $\frac{1}{n}\sum f(1+m/n)$ is the total area of n thin rectangles of width $\frac{1}{n}$ standing on $[1,2]$, and as the rectangles get thinner they fill out the area under the curve exactly. The other two limits are this same picture in disguise: the job is to pull out a factor $\frac{1}{n}$ and write each term as a function of m/n alone.

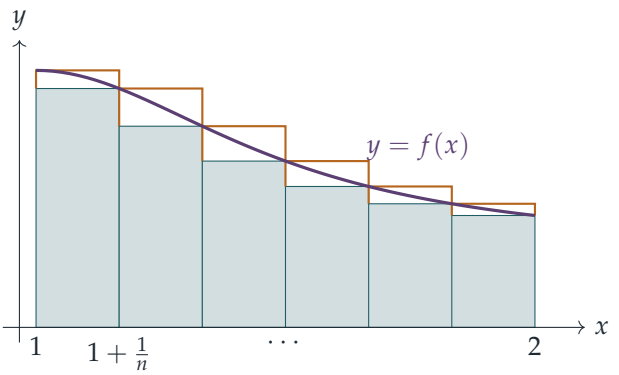
Solution

The limit formula

Split $[1,2]$ into n strips of width $\frac{1}{n}$; the mth strip is $[1+\frac{m-1}{n},\, 1+\frac{m}{n}]$. On it, stand a rectangle whose height is the value of f at the strip's right-hand end, $f(1+m/n)$. Its area is $\frac{1}{n}f(1+m/n)$, so

$$S_n = \frac{1}{n}\sum_{m=1}^{n} f(1+m/n)$$

is the total area of the rectangles.



The sketch shows a decreasing f. The shaded (teal) rectangles use right-hand heights

and so lie *under* the curve; the amber steps on top show the rectangles we would get using left-hand heights, which lie *over* it. Hence

$$S_n \;\le\; \int_1^2 f(x)\,\mathrm{d}x \;\le\; S_n + (\text{total area of the amber pieces}).$$

Now slide all the amber pieces sideways into a single column of width $\frac{1}{n}$: they stack up to a total height of $f(1) - f(2)$, because the heights telescope. So

$$0 \;\le\; \int_1^2 f(x)\,\mathrm{d}x - S_n \;\le\; \frac{f(1)-f(2)}{n} \;\longrightarrow\; 0 \quad \text{as } n \to \infty,$$

and $S_n \to \int_1^2 f(x)\,\mathrm{d}x$. An increasing f works the same way with the inequalities reversed, and any reasonable (say continuous, piecewise monotonic) function can be cut into finitely many monotonic pieces, each contributing an error of order $\frac{1}{n}$. This justifies

$$\lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} f(1 + m/n) = \int_1^2 f(x)\,\mathrm{d}x.$$

The limit $\ln 2$

Ask yourself How do we make $\dfrac{1}{n+m}$ look like $\dfrac{1}{n} f(1+m/n)$?

Take out a factor n from the denominator:

$$\frac{1}{n+m} = \frac{1}{n} \cdot \frac{1}{1+m/n} = \frac{1}{n} f(1+m/n) \qquad \text{with } f(x) = \frac{1}{x}.$$

So

$$\begin{aligned} \lim_{n\to\infty} \left(\frac{1}{n+1} + \cdots + \frac{1}{n+n} \right) &= \lim_{n\to\infty} \frac{1}{n} \sum_{m=1}^{n} \frac{1}{1+m/n} \\ &= \int_1^2 \frac{1}{x}\,\mathrm{d}x = \ln 2 - \ln 1 = \ln 2, \end{aligned}$$

as required. (Here f is decreasing, so the sketch argument applies exactly; it even tells us the sum lies between $\ln 2 - \frac{1}{2n}$ and $\ln 2$.)

The final limit

The mth term is $\dfrac{n}{n^2+m^2}$, for $m = 1, \ldots, n$. Divide top and bottom by n^2:

$$\frac{n}{n^2+m^2} = \frac{1}{n} \cdot \frac{1}{1+(m/n)^2}.$$

To match the formula, which evaluates f at $x = 1 + m/n$, we need $f(x) = \dfrac{1}{1+(x-1)^2}$, since then $f(1+m/n) = \dfrac{1}{1+(m/n)^2}$. This f is decreasing on $[1,2]$. Therefore

$$\lim_{n\to\infty} \sum_{m=1}^{n} \frac{n}{n^2+m^2} = \int_1^2 \frac{\mathrm{d}x}{1+(x-1)^2} = \Big[\tan^{-1}(x-1)\Big]_1^2$$
$$= \tan^{-1} 1 - \tan^{-1} 0 = \frac{\pi}{4}.$$

(Equivalently, substitute $u = x - 1$ to get $\int_0^1 \frac{\mathrm{d}u}{1+u^2}$.)

Checks

The sums can be computed directly. With $n = 1000$, the second sum is 0.69290 against $\ln 2 \approx 0.69315$, and the third is 0.78515 against $\pi/4 \approx 0.78540$. In both cases the shortfall is about $\frac{1}{4000}$, inside the bound $\frac{f(1)-f(2)}{n} = \frac{1}{2000}$ from the sketch. Both sums also come in slightly *below* the limit, as they should for a decreasing f with right-hand heights.

Common slips

- Writing the third limit as $\int_1^2 \frac{\mathrm{d}x}{1+x^2}$: the formula evaluates f at $1 + m/n$, not at m/n, so the shift by 1 must go into f.
- Forgetting that the last term ($m = n$) is $\frac{n}{2n^2}$, which matches $\frac{1}{n}f(2)$: the sum runs over all n right-hand endpoints, not $n + 1$ points.

Beyond STEP

These are Riemann sums, and the sketch argument is the seed of the definition of the Riemann integral: a function is integrable when the upper and lower sums can be squeezed together, and every continuous function on a closed interval has this property. The same idea also runs backwards, turning sums into integrals to estimate them, as in the proof that $1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n$ converges. Abbott's *Understanding Analysis* has a very readable chapter on the Riemann integral.

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